

ANTÓN GÓMEZ LÓPEZ

Artificial Intelligence - Computer Engineer - Mathematics

 [antongomez.github.io/portfolio](https://github.com/antongomez)

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SUMMARY

Motivated Computer Engineer specializing in AI, with hands-on experience in deep learning model development and research currently in preparation for peer-reviewed publication. Recipient of multiple academic awards (including the End-of-Degree Award) and research scholarships. Demonstrated success adapting StyleGAN3 for remote sensing, achieving Average Accuracy within 1 percentage point of state-of-the-art ViT models. Contributor to the open-source project caretEnsemble, with strong expertise in Pytorch. Additionally, I have academic and small-project experience with C++, Java, and JavaScript, some frontend experience with React, and regularly use tools like Git and Docker. Naturally curious and continuously expanding skills through independent projects.

EDUCATION

- 09/24 - 02/26 **Master's Degree in Artificial Intelligence** | [Link to all courses](#) **University of Santiago de Compostela**
- Relevant Courses: Natural Language Understanding (TensorFlow), Machine Learning, Computer Vision (TensorFlow), Intelligent Robotics, AI Project Management, Deep Learning (Tensorflow), AI in Big Data Environments (PySpark)
 - Grade: 9.28/10
 - Courses finished with honours: 3
 - **Final Thesis:** *Adaptation of StyleGAN for Processing Multidimensional Remote Sensing Images*
 - Adapting StyleGAN3 to classify drone-acquired multispectral images from highly imbalanced and low-sample datasets, achieving Average Accuracy (AA) within 2.5 percentage point of state-of-the-art ViT models, by extending the discriminator with a fully connected classification layer and training the model from scratch on multispectral data.
 - **Publication:** Manuscript in preparation for a top-tier peer-reviewed journal.
- 09/18 - 07/24 **Bachelor's Degree in Computer Engineering** | [Link to all courses](#) **University of Santiago de Compostela**
- Relevant Courses: Algorithms and Data Structures (programming language C), Computer Architecture, Object Oriented Programming (Java), Formal Languages and Automata Theory, Intelligent Systems (Python), Cybersecurity
 - Grade: 9.24/10
 - Courses finished with honours: 9
 - **Final Thesis:** *Effective Adaptation of GANs for Multidimensional Remote Sensing Images* — Initial StyleGAN2 adaptation later extended in Master's research. [Project Landing Page](#).
- 09/18 - 07/24 **Bachelor's Degree in Mathematics** | [Link to all courses](#) **University of Santiago de Compostela**
- Relevant Courses: Linear and Multilinear Algebra, Numerical Methods in Optimization and Differential Equations, General Topology, Probability and Statistics, Regression Models and Multivariate Analysis (programming language R)
 - Grade: 8.94/10
 - Courses finished with honours: 8
 - **Final Thesis:** *Classification Methods and Classifier Ensembles in Supervised Learning*.

EXPERIENCE

- 02/25 - 02/26 **AI Research Scholar, Remote Sensing** **Research Center in Intelligent Technologies (CiTIUS)**
scholarship holder
- Redesigned StyleGAN3 for complex classification tasks, targeting severe data scarcity and class imbalance in drone-acquired multispectral imagery.
 - Modified the network discriminator by integrating a fully connected classification head, executing end-to-end training from scratch on multidimensional datasets.
 - Achieving Average Accuracy (AA) within 2.5 percentage point of state-of-the-art ViT models.
- 09/23 - 12/23 **Research Assistant, Statistical Classification** **Galician Centre for Mathematical Research and Technology**
- Architected and implemented multiclass support for the open-source *caretEnsemble* R package, extending the existing codebase to successfully integrate the team's novel statistical research.
 - Collaborated cross-functionally to translate mathematical models into robust, tested software features for the open-source community.
- 06/23 - 08/23 **Intern, Machine Learning** **Gradiant, Galician R&D Center in Advanced Telecommunications**
- Optimized NLP inference, boosting system performance by 2.5x through parallelization techniques (DDP).
 - Benchmarked and integrated distributed computing frameworks (Ray, Dask) improving system scalability.

07/22 – 08/22
scholarship holder

Intern, Natural Language Processing

Research Center in Intelligent Technologies (CiTIUS)

- Studied neural machine translation architectures and key techniques within the NLP group, focusing on low-resource language scenarios.
- Trained and evaluated a Galician–Spanish translation model using OpenNMT, achieving a BLEU score of 64.3%.

AWARDS

05/25	End-of-Degree Award	Government of Galicia
	Selected as one of 4 recipients among over 300 Computer Engineering students in Galicia, Spain, for having one of the top academic records in the 2024 graduating cohort.	
11/24	XX Awards for Linguistic Quality in Galician for Final Degree Projects	University of Santiago de Compostela
	Selected as the best Computer Engineering bachelor's thesis in Galician in the Engineering and Architecture area at the University of Santiago de Compostela in 2024 for excellence in linguistic quality in the Galician language.	
09/24	XVII Award for the Best Free Software University Project	Interuniversity Consortium of Galicia (CiXUG)
	Awarded second prize for the Best Free Software University Project among all free software university projects submitted in Galicia in 2024 by the Interuniversity Consortium of Galicia for adopting free software principles in my bachelor's thesis.	

PROJECTS & PROGRAMMING COMPETITIONS

2024	Contribution to R package <i>caretEnsemble</i>	github.com/zachmayer/caretEnsemble
	• Contributed to the <i>caretEnsemble</i> R package, adding multiclass support and novel ensemble capabilities. • Engaged in collaborative open-source development, applying professional software engineering practices including version control (Git), unit testing, code reviews, and pull requests.	
2023 – 2025	Programming competitions	Ada Byron, HackUDC, HackUPC
	• Participated in national university competitions and hackathons, including Ada Byron and HackUPC. • Learned new programming languages (Python, C++, JavaScript, Rust) and solved challenges beyond the academic curriculum.	
2024 – 2025	Professional website development	alvaroleonnutricion.com
	• Developed a responsive professional website using React, focused on simplicity and user-friendly design. • Gained experience in front-end development best practices, including component-based architecture, state management, and responsive design.	

LANGUAGES

English - C1 Advanced (Cambridge Certificate), **Spanish** - native, **Galician** - native

07/22 - 08/22	English course	Limerick Language Center
	A three-week stay in Limerick taking a B2-level English.	

VOLUNTEERING

02/26	Hackathon Organizer	A Coruña, Galicia, Spain
	Served as an organizer of HackUDC 2026, the second largest university hackathon in Spain (over 400 participants, €8k in prizes), coordinating sponsors, defining challenges and prizes, and ensuring educational value for participants.	
07/21	Environmental volunteering	Baixa Limia Natural Park, Galicia, Spain
	Performed cleaning tasks, adapted areas, and monitored native forest regeneration zones.	
06/21 - 09/21	Cultural volunteering	Cambre, Galicia, Spain
	Promoted the English Way through a volunteer initiative supported by the Cambre City Council, providing information about local heritage and assisting pilgrims.	
06/19 - 09/19	Tourist Entertainer	Praga, Czech Republic
	Collaborated with a hostel in Prague (Hostel Advantage) as a volunteer, performing daily tasks at the hostel and organizing entertainment and dynamization activities within the hostel.	